

Element D

The Beauty and the Beast rose must be engineered in a very comprehensive and precise way, calling for the most pristine designing, analysis, and selection processing. Because of this, several steps were taken to ensure the most efficient and effective solution.

First, a list of criteria was made over the span of several days based largely on the suggestions Mrs. Moody, the theater director, who answered questions about the rose project and gave us this list of requirements:

- Add light so the rose illuminated in a way
- Rose must stay on the pedestal the whole time
- The pedestal will be on a wagon which will move on and off the stage
- wants remote control
- One rose has wilt and two roses lost even more.
- Needs to have 3 maybe 4 petals to fall
- Rose falling should be silent or audience shouldn't hear it
- Must have 13" closure
- There are two dress rehearsals. Feb 19-22. needed by tech week. 4 times. five performances. used 10 times
- Must be a warm light
- The rose must be red
- Size range is a foot and a half
- The drop doesn't matter
- The bulb of rose must face the audience
- Must keep its cylindrical shape

After two days of amalgamating issues and requirements, the criteria list shown in Element C was created.

Next we searched the web and researched other designs for the rose and recorded our findings. Using these ideas, our own critiques, and two days of brainstorming, we compiled a list of ideas for the rose design. We placed these designs in a decision matrix and ranked each 1-10 for how well the designs fit each criteria (1 being the lowest). After adding the score for each design up, it was discovered that the string design fit the requirements the best.

The vague design goes as follows.

The stem is made of sturdy unbendable materials that support 3-5 petals that are. These petals are not attached to the stem but held up by strings running through the stem and connected on the other end to spools in the base. A motor will turn each spool, letting out more string for the petals and causing the petals to fall due to gravity. The string method is great for slowing the fall of the petals to make it more noticeable to the audience because the motors can control the speed of the fall of petals. The motors will be controlled by remote controllers that someone on or offstage would operate. The stem will be bent slightly so that the rose bud that rests on the tip of the stem will face towards the audience and not simply straight up. The petals will be painted red, the stem will be painted green, and the base that hides the motors and spools will be painted brown.